RetailIQ: An End-to-End E-Commerce Customer and Sales Analytics Platform

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***Abstract*—E-commerce platforms generate large volumes of customer, order, product, payment, return and review data, which makes it difficult to obtain a unified view of business performance and customer behaviour. This paper presents RetailIQ, an end-to-end e-commerce analytics platform that integrates data preprocessing, feature engineering, exploratory data analysis, customer, product, sales and geographical analysis, customer segmentation and interactive visualization in a single reproducible workflow. Seven relational tables from a simulated Indian marketplace (25,000 customers, 100,000 orders, 254,331 order items, 550 products, 100,000 payments, 12,075 returns and 77,530 reviews) are cleaned and integrated into a master table of 254,331 rows, enriched with 15 engineered temporal, financial, customer, product, discount and return features. Customer behaviour is modelled with Recency, Frequency and Monetary (RFM) quintile scoring, which places 22,820 purchasing customers into six actionable segments, and with K-Means clustering on standardized RFM features, where the silhouette coefficient selects K = 2 (s = 0.4378). The analysis shows that Electronics leads revenue (₹122.36 Cr) at only a 0.69% margin while Fashion is the most profitable category (33.14% margin), that 33 products are loss-making, that the overall return rate is 12.07% with size and damage issues as the leading causes, that UPI is the dominant payment method, and that 59.7% of reviews are positive. All results are delivered through a seven-page interactive Streamlit dashboard, demonstrating how integrated analytics can support data-driven retail decision-making.**

Keywords*—*e-commerce analytics, customer segmentation, RFM analysis, K-Means clustering, silhouette coefficient, sales analytics, business intelligence dashboard, Streamlit

I. Introduction

A. Background

The rapid growth of online retail in India has produced very large volumes of operational data. Every purchase leaves a trail across several systems: the customer profile, the order header, the individual order lines, the product catalogue, the payment gateway, the returns desk and the review platform. Each source holds valuable information about purchasing behaviour, product performance, sales trends and revenue, yet the sources are usually stored and analysed in isolation.

Analysing these sources one at a time hides the business as a whole. A category may look healthy on revenue but be barely profitable; a customer may look valuable on total spend but has not ordered in two years; a product may sell well but generate a disproportionate share of returns. Such insights only appear when the data is joined, cleaned and examined together, and manual spreadsheet analysis of hundreds of thousands of rows is slow and error-prone [1], [2].

B. Problem Statement

The problem addressed in this work is the development of an integrated e-commerce analytics platform that can (i) preprocess and integrate heterogeneous customer, order, product, payment, return and review data; (ii) identify customer purchasing patterns and valuable or at-risk customer segments; (iii) compare product and category performance on revenue, profit, margin and rating simultaneously; (iv) reveal temporal and geographical sales trends; and (v) present the resulting insights through a centralized, interactive interface for business users.

C. Research Objectives

1) To preprocess and integrate data from the customer, order, order-item, product, payment, return and review sources into one analysis-ready dataset.
2) To analyse customer purchasing behaviour and segment customers using RFM analysis and K-Means clustering.
3) To analyse product and category performance using revenue, profit, margin, inventory and rating measures.
4) To analyse sales and revenue trends across time and Indian states.
5) To analyse operational indicators: returns, payment methods and customer review sentiment.
6) To develop an interactive dashboard that presents the key performance indicators and insights in one place.

D. Research Questions

RQ1: How can heterogeneous e-commerce data be effectively preprocessed and integrated to support comprehensive business analysis?

RQ2: Which customer purchasing patterns and segments can be identified using RFM analysis and clustering?

RQ3: How do product, category, sales, revenue and geographical data reveal meaningful business performance patterns?

RQ4: How can an interactive dashboard present integrated e-commerce analytics and key insights in a centralized interface?

E. Contributions

- **Integrated analytics workflow** that combines seven relational sources into a single reproducible pipeline with 100% referential integrity.
- **Cross-domain feature engineering** of 15 temporal, financial, customer, product, discount and return attributes.
- **Dual customer segmentation** using rule-based RFM scoring and silhouette-validated K-Means clustering.
- **Multi-dimensional business analysis** of category profitability, loss-making products, regional performance, returns, payments and reviews.
- **A seven-page interactive Streamlit dashboard** that maps each analysis to a stakeholder group.

F. Organisation of the Paper

Section II reviews related work. Section III describes the proposed methodology, Section IV the experimental setup and Section V the results. Section VI discusses the findings, Section VII states the limitations, and Section VIII concludes the paper with directions for future work.

II. Related Work

A. E-Commerce and Retail Analytics

Data-driven decision making is foundational to modern retail. Ngai *et al.* [1] provided a widely cited classification of data-mining techniques for Customer Relationship Management (CRM) along customer identification, attraction, retention and development. Chen *et al.* [2] applied RFM-based segmentation and decision trees to an online retailer and showed that transaction logs alone can yield actionable customer groups. Griva *et al.* [3] segmented retail customer visits using market-basket data, while Agrawal and Srikant [4] introduced the association-rule mining algorithms that underpin basket analysis. These studies, however, typically address one analytical dimension at a time.

B. Customer Behaviour and RFM Analysis

Recency, Frequency and Monetary (RFM) analysis, popularised by Hughes [5], remains one of the most practical models for evaluating customer value from purchase history. Cheng and Chen [6] combined RFM attributes with rough-set theory to classify customer value, and Birant [7] demonstrated RFM-based segmentation for targeted marketing. RFM compresses high-dimensional transaction data into three interpretable behavioural variables, which makes it well suited to business users. RetailIQ adopts quintile-based RFM scoring to group 22,820 purchasing customers into six actionable segments.

C. Unsupervised Clustering

K-Means, introduced by MacQueen [8] and reviewed by Jain [9], is the most widely used algorithm for discovering natural groupings in customer data. Because K-Means is distance-based, feature scaling is essential when monetary values dominate count variables. Rousseeuw [10] proposed the silhouette coefficient to measure cluster cohesion and separation, and Kaufman and Rousseeuw [11] established it as a standard internal validation index for choosing the number of clusters. Alternatives such as BIRCH [12] and CURE [13] target very large or non-spherical data, but K-Means remains favoured in business segmentation for its interpretability and efficiency [14]. RetailIQ applies K-Means to standardized RFM features and selects K from 2 to 7 by the silhouette coefficient.

D. Business Intelligence and Dashboards

Decision-support systems translate analytical results into managerial action [15]. Few [16] established visual design principles for dashboards that emphasise information density, hierarchy and at-a-glance clarity. Open-source Python tools such as Pandas [17], scikit-learn [18], Plotly [19] and Streamlit [20] now make it possible to build interactive analytical web applications without a separate front-end stack [21]. RetailIQ uses this stack to deliver a multi-page dashboard directly from the analysis code.

E. Critical Comparison

Table I compares representative approaches. RFM and clustering studies provide strong customer segmentation but rarely connect segments to product profitability or operations. Market-basket studies reveal co-purchase patterns but ignore customer value and returns. Dashboard design work offers presentation principles but not the analytical pipeline that feeds the dashboard.

TABLE I
COMPARISON OF RELATED APPROACHES

Approach	Main focus	Limitation
RFM / rule-based [2], [7], [5], [6]	Customer value tiers	Customer view only
K-Means + silhouette [8], [9], [10], [11]	Data-driven segments	No product or operations link
Market basket [3], [4]	Co-purchase rules	Ignores value and returns
BI dashboards [15], [16]	Visual decision support	No analytics pipeline
RetailIQ (this work)	Customer + product + sales + operations	Simulated data, batch mode

F. Research Gap

Existing work addresses individual analytical domains – standalone RFM segmentation, K-Means clustering, basket analysis or dashboard design – but rarely presents a fully integrated, reproducible framework. RetailIQ bridges this gap by combining multi-source preprocessing, cross-domain feature engineering, dual customer segmentation, category profitability analysis, regional analysis, return and payment tracking, review sentiment and interactive visualization within one workflow.

III. Proposed Methodology

A. Overall Framework Architecture

RetailIQ is organised as five stacked layers (Fig. 1): a data-source layer holding the seven raw tables; a processing layer for inspection, cleaning, integration and feature engineering; an analytics layer for EDA, RFM, K-Means, product, sales, geography, returns, payment and review analysis; a visualization layer implemented as a multi-page Streamlit dashboard; and the end users. Data flows strictly downward, and each layer communicates with the next only through files, which makes every stage independently re-runnable. The ten-stage implementation workflow is shown in Fig. 2.

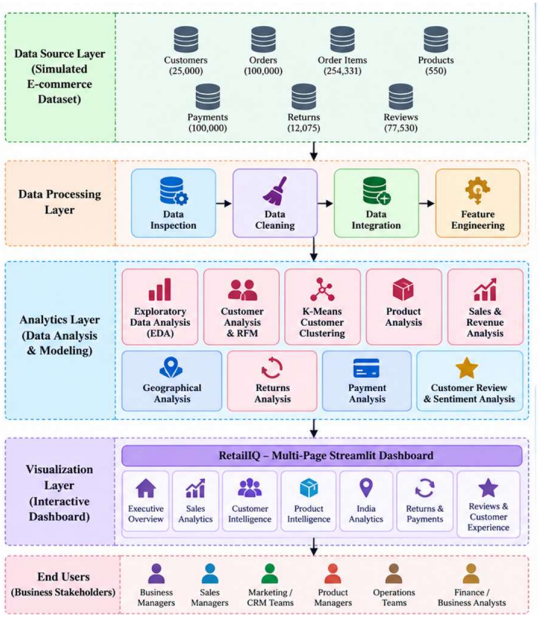


Fig. 1. Layered system architecture of RetailIQ.

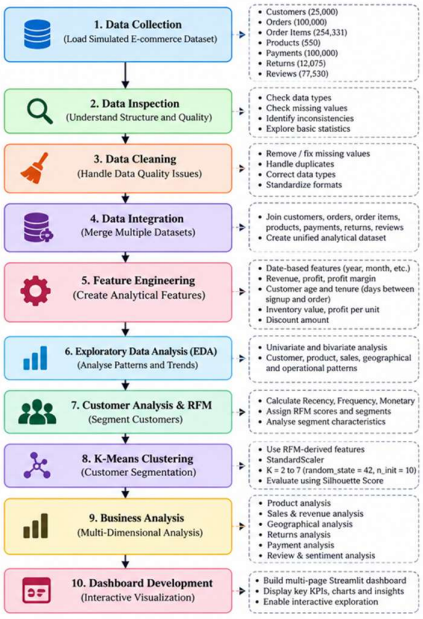


Fig. 2. Ten-stage implementation workflow of RetailIQ.

B. Dataset Description

The study uses a simulated Indian e-commerce dataset [22] of seven relational tables that form a star-like schema centred on orders (Table II). Customers place orders; each order contains one or more order items, each referring to one product; every order has a payment record and may have returns; reviews link a customer, an order and a product.

TABLE II
DATASET SCHEMA AND RECORD COUNTS

Table	Records	Key attributes
customers	25,000	Age, gender, state, city, signup date, loyalty tier
orders	100,000	Customer, date, status, coupon, campaign
order_items	254,331	Order, product, quantity, price, discount
products	550	Category (22), brand (148), cost, stock, rating
payments	100,000	Method, status, amount, refund
returns	12,075	Order, reason, refund status
reviews	77,530	Product, customer, rating, sentiment, verified

C. Data Preprocessing and Quality Analysis

Profiling scripts checked shapes, data types, distinct values and referential integrity, following standard data-preparation practice [23], [24]; no orphan foreign keys were found in any relation. All date fields were converted to datetime objects, and each missing-value pattern was treated according to its meaning (Table III). Missing sentiment labels were derived deterministically from star ratings (4–5 Positive, 3 Neutral, 1–2 Negative), a common rating-based proxy for review polarity [25], and missing last-order dates of customers who never purchased were intentionally kept as null to preserve that behavioural state.

TABLE III
MISSING-VALUE TREATMENT

Column	Missing	Treatment
customers.age	250	Median age (37 years)
customers.preferred_device	125	"Unknown"
customers.last_order_date	2,300	Kept null (never purchased)
orders.coupon_code	33,711	"No Coupon"
orders.campaign_id	82,821	"No Campaign"
products.rating_average	4	Median rating
reviews.review_sentiment	233	Derived from rating

D. Data Integration and Feature Engineering

The master analytical table was built with three left joins: orders → customers on customer_id, → order_items on order_id, and → products on product_id, giving one row per order line (254,331 rows × 54 columns). Returns were

attached as a binary flag rather than a join key to avoid one-to-many row duplication; the flag marks 25,739 order lines belonging to 9,510 orders. Fifteen features were then engineered, bringing the table to 69 columns:

- **Temporal:** order year, month, month name, quarter, day and day name.
- **Financial:** item_profit = revenue − cost and profit_margin = item_profit / revenue × 100, with infinite values set to 0.
- **Customer:** age and customer_lifetime_days = order_date − signup_date.
- **Product:** inventory_value = stock × cost_price and profit_per_unit.
- **Discount and returns:** discount_amount = unit_price × quantity × discount% / 100, and has_return.

E. RFM Customer Segmentation

For each of the 22,820 customers with at least one order, three metrics were computed relative to the latest order date in the data (31 December 2025): Recency *R* (days since the last order), Frequency *F* (number of distinct orders) and Monetary value *M* (total revenue). Each metric was scored from 1 to 5 by quintiles, with recency reversed so that recent buyers score high. Segments were assigned by first-matching rules: Champions ($R \geq 4$, $F \geq 4$, $M \geq 4$), Loyal Customers ($R \geq 3$, $F \geq 3$), New Customers ($R \geq 4$, $F \leq 2$), At Risk ($R \leq 2$, $F \geq 3$), High Value At Risk ($R \leq 2$, $M \geq 4$) and Regular Customers (all others).

F. K-Means Clustering

To complement the rule-based segments with a data-driven grouping, the R, F and M variables were standardized to zero mean and unit variance so that rupee amounts in lakhs do not dominate order counts. K-Means models were trained for K = 2 to 7, and the model with the highest mean silhouette coefficient was selected. For a sample *i*, the silhouette is

$$s(i) = [\, b(i) - a(i) \,] \,/\, \max\{\, a(i),\, b(i) \,\} \qquad (1)$$

where $a(i)$ is the mean distance from i to the other members of its cluster and $b(i)$ is the mean distance to the members of the nearest other cluster; $s(i)$ lies in $[-1, 1]$, and higher values indicate better-defined clusters [10].

G. Business Analysis Modules and Dashboard

Product analysis aggregates quantity, revenue, profit, margin, rating and stock by product, category and brand; geographical analysis aggregates orders, customers, revenue and margin by state and city; operations analysis summarises returns by reason, status and category, payments by method and status, and reviews by rating, sentiment and verification. All outputs are pre-computed and loaded once into a cached Streamlit application whose seven pages – Executive Overview, Sales Analytics, Customer Intelligence, Product Intelligence, India Analytics, Returns & Payments, and Reviews & Customer Experience – each serve a specific stakeholder group.

IV. Experimental Setup

A. Software Environment

The pipeline and dashboard were implemented in Python 3.13 on 64-bit Windows using Pandas 2.2.3 and NumPy 2.1.2 for data manipulation, scikit-learn 1.6.1 for scaling, K-Means and silhouette computation, Matplotlib 3.10.0 and Plotly 6.0.0 for static and interactive charts, and Streamlit 1.42.0 for the web dashboard.

B. Hardware Environment

Experiments were run on a workstation with a multi-core x86_64 processor at 2.50 GHz, 16 GB RAM and solid-state storage. The full pipeline, including clustering on 22,820 customers, executes locally within minutes.

C. Model Configuration

K-Means used StandardScaler-normalised RFM features, random_state = 42, n_init = 10, a maximum of 300 iterations and a convergence tolerance of 10^{-4}. Cluster counts from K = 2 to K = 7 were evaluated sequentially.

D. Evaluation Metrics

Clustering quality was measured by the mean silhouette coefficient (1). Business performance was measured by profit margin = total profit / total revenue × 100, return rate = returns / orders × 100, and the percentage of positive, neutral and negative reviews. Pipeline correctness was verified by row-count reconciliation between stages and by checking that total revenue is identical across the category, state and dashboard KPI views.

V. Experimental Results

A. Data Integration and Overall Performance

The pipeline joined all seven tables into a 254,331 × 69 feature table with zero orphan records. Across January 2023 to December 2025 the marketplace generated ₹244.99 Cr of revenue and ₹19.73 Cr of net profit (8.1% margin), with total discounts of ₹36.29 Cr. Revenue was roughly flat in 2023 and 2024 (₹66.1 Cr and ₹65.2 Cr) and rose sharply to ₹113.7 Cr in 2025, peaking in the final quarter (Fig. 3). Weekends earn about 25% more revenue per day than weekdays. Of all orders, 80.7% were delivered, 8.1% cancelled, 6.7% returned, 2.9% failed and 1.7% were still processing.



Fig. 3. Monthly revenue and profit, January 2023 – December 2025.

B. RFM Segmentation Results

Table IV and Fig. 4 report the six RFM segments. Champions (13.7% of buyers) order 8.4 times on average and last ordered 41 days before the snapshot, contributing about 31% of total customer spend. At Risk customers, who used to buy six times on average but have been silent for almost two years, hold another 31% of spend, making them the largest win-back opportunity.

TABLE IV
RFM SEGMENT PROFILE (22,820 CUSTOMERS)

Segment	Cust.	Share	Avg R (days)	Avg F	Avg M (₹)
Regular	5,693	24.9%	575.5	1.6	25,575
Loyal	5,628	24.7%	160.7	5.1	1,05,797
At Risk	4,945	21.7%	710.4	6.0	1,51,822
Champions	3,119	13.7%	41.0	8.4	2,47,100
New	2,967	13.0%	57.4	1.8	38,930
High Value At Risk	468	2.1%	732.9	1.9	1,53,651

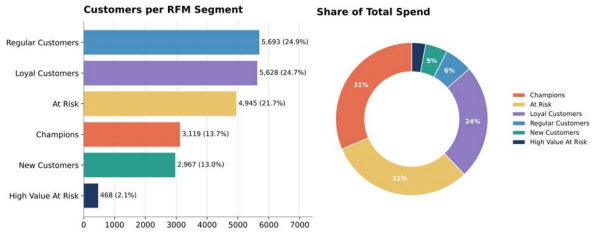


Fig. 4. Customers per RFM segment (left) and share of total spend (right).

C. K-Means Clustering Results

Table V lists the silhouette coefficient for each K. K = 2 achieves the highest score (0.4378), followed by K = 3 (0.4107), while K = 6 and K = 7 show clear overlap (Fig. 5). The two selected clusters (Table VI, Fig. 6) separate 17,284 Core Moderate Shoppers from 5,536 High-Value Frequent Shoppers who order about three times as often and spend about five times as much.

TABLE V
SILHOUETTE COEFFICIENT FOR K = 2 TO 7

K	2	3	4	5	6	7
s	0.4378	0.4107	0.3983	0.4013	0.3338	0.3419

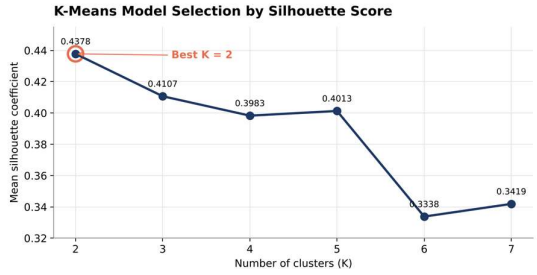


Fig. 5. Silhouette score versus number of clusters; K = 2 is optimal.

TABLE VI
K-MEANS CLUSTER PROFILE (K = 2)

Cluster	Customers	Avg R (days)	Avg F	Avg M (₹)
0 – Core Moderate	17,284	387.7	2.90	53,417
1 – High-Value Frequent	5,536	295.0	9.01	2,75,766

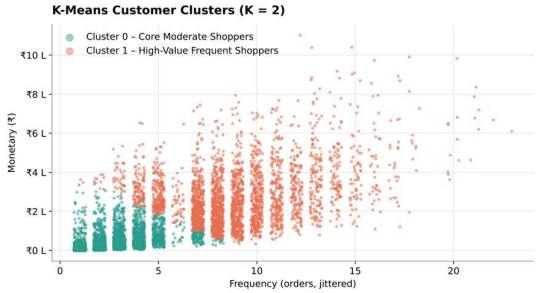


Fig. 6. Frequency versus monetary value of the two K-Means clusters.

D. Product and Category Profitability

Category analysis reveals a large gap between revenue and profitability (Table VII, Fig. 7). Electronics generates the highest revenue (₹122.36 Cr from 59,068 units) but only a 0.69% margin, whereas Fashion earns the highest net profit (₹4.62 Cr) at a 33.14% margin from just ₹13.94 Cr of revenue. Footwear and Bags also exceed 28% margin. At product level, 33 of the 550 products have negative total profit and are flagged for pricing or sourcing review.

TABLE VII
PERFORMANCE OF THE TOP REVENUE CATEGORIES

Category	Units	Revenue (₹ Cr)	Profit (₹ Cr)	Margin	Rating
Electronics	59,068	122.36	0.85	0.69%	3.97
Appliances	35,842	40.52	1.57	3.89%	4.00
Home & Kitchen	60,205	15.81	2.84	17.96 %	3.92
Furniture	8,118	14.70	1.43	9.76%	3.87
Fashion	98,750	13.94	4.62	33.14 %	4.01
Footwear	32,501	8.26	2.43	29.42 %	4.04
Bags	29,948	5.59	1.60	28.59 %	4.20

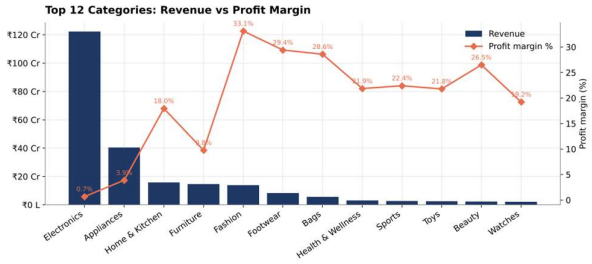


Fig. 7. Revenue (bars) and profit margin (line) of the top 12 categories.

E. Geographical Analysis

Maharashtra leads with ₹34.27 Cr of revenue from 14,196 orders, followed by Uttar Pradesh (₹26.64 Cr) and Gujarat (₹25.95 Cr) (Fig. 8). State-level margins are tightly clustered between about 7.6% and 8.8%, so regional differences are driven by order volume rather than by profitability.

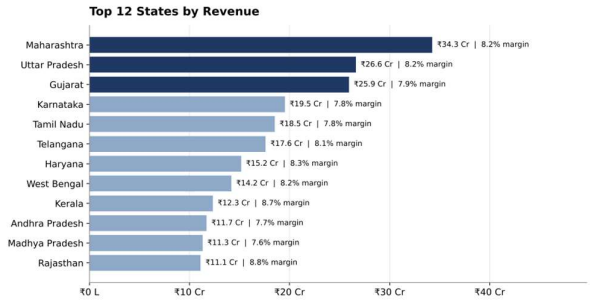


Fig. 8. Top 12 states by revenue with profit margin.

F. Returns and Payments

There were 12,075 returns against 100,000 orders, an overall return rate of 12.07%. The leading reasons are size issues (2,201; 18.2%), change of mind (1,984; 16.4%), damaged products (1,933; 16.0%) and quality issues (1,692; 14.0%) (Fig. 9); size and damage together explain 34.2% of all returns. Fashion carries the most returns (3,185). Refunds are completed for 8,473 returns, approved for 1,792, rejected for 938 and pending for 872.

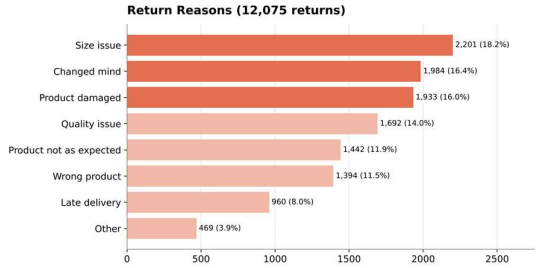


Fig. 9. Distribution of return reasons (12,075 returns).

UPI is the dominant payment method with 31,971 transactions (₹78.06 Cr), followed by Cash on Delivery (17,274), credit cards (15,063) and debit cards (12,682). Overall, 82.1% of payments succeed, 10.7% are refunded and 5.1% fail (Fig. 10).

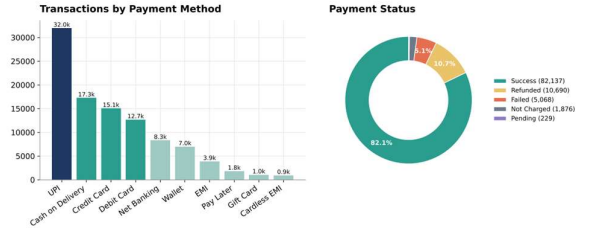


Fig. 10. Transactions by payment method (left) and payment status (right).

G. Customer Reviews and Sentiment

Of 77,530 reviews, 46,311 (59.7%) are positive, 24,208 (31.2%) neutral and 7,011 (9.0%) negative, and 97.1% come from verified purchases (Fig. 11). Four-star ratings are the most common (31,265). Pet Supplies (4.40) and Toys (4.36) have the highest average category ratings.

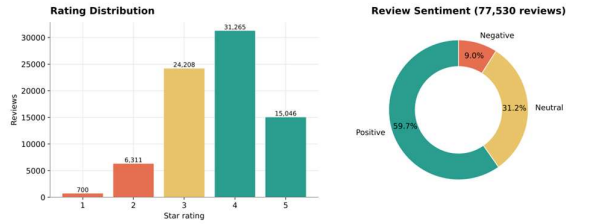


Fig. 11. Rating distribution (left) and review sentiment split (right).

H. Interactive Dashboard

All analytical outputs are integrated into a seven-page Streamlit dashboard. The Executive Overview page (Fig. 12) summarises revenue, profit, orders, purchasing customers and return rate in KPI tiles, with a revenue trend, category ranking and order-status donut. The remaining pages provide date and category filters for sales, RFM and cluster views with segment recommendations, category and brand profitability with the loss-making product list, state and city rankings, return and payment breakdowns, and rating and sentiment analysis. Because data loading is cached, page switches and filter changes are served from memory in under a second.

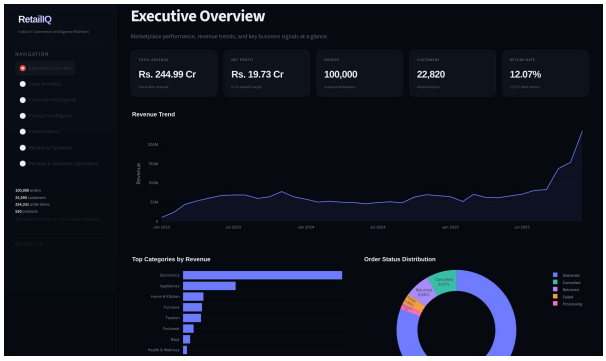


Fig. 12. Executive Overview page of the RetailIQ dashboard.

VI. Discussion

A. Interpretation of Findings

The results answer the four research questions. For RQ1, the file-based pipeline shows that seven heterogeneous tables can be integrated without row loss or duplication when one-

to-many relations such as returns are represented as flags. For RQ2, customer value is highly concentrated: Champions and Loyal Customers form 38.4% of buyers but more than half of total spend, while At Risk and High Value At Risk customers (23.8% of buyers) hold about a third of spend that is vulnerable to churn. K-Means independently confirms a small high-value group, although with K = 2 the clusters are broad. For RQ3, revenue is not profit: Electronics dominates revenue at a 0.69% margin, while Fashion, Footwear and Bags generate far higher margins, so promotion decisions based on revenue alone would favour the wrong categories. Returns have fixable operational causes – better size charts and protective packaging address over a third of returns. For RQ4, the dashboard places all these views in one interface, turning descriptive analytics into decisions that each business function can act on [26].

B. Stakeholder Decision Support

- **Executive leadership** monitors revenue, profit, margin and return rate on the Executive Overview.
- **Marketing and CRM teams** target At Risk and High Value At Risk customers with win-back campaigns.
- **Merchandising teams** review the 33 loss-making products and shift promotion toward high-margin categories.
- **Operations teams** track return reasons and refund status to reduce size and damage issues.
- **Finance teams** monitor payment-method adoption, failure and refund rates and discount erosion.

VII. Limitations

1) **Simulated data:** findings such as category margins and return rates reflect the simulated dataset and cannot be generalised to specific real platforms.
2) **Batch processing:** the pipeline works on a fixed snapshot, and new data requires re-running the scripts.
3) **Clustering sensitivity:** K-Means results depend on feature scaling and K and must be re-validated as behaviour changes.
4) **Rule-based sentiment:** sentiment is derived from star ratings rather than from review text.
5) **No access control:** the dashboard is intended for local or trusted use only.

VIII. Conclusion and Future Work

A. Conclusion

This paper presented RetailIQ, an integrated end-to-end e-commerce analytics framework and interactive business-intelligence dashboard. The platform cleans and integrates seven heterogeneous tables covering 25,000 customers, 100,000 orders, 254,331 order items, 550 products, 100,000 payments, 12,075 returns and 77,530 reviews into a single 69-column analytical table. RFM scoring segmented 22,820 purchasing customers into six actionable tiers, and silhouette-validated K-Means identified two behavioural clusters (K = 2, s = 0.4378). Profitability analysis exposed the gap between revenue and margin and identified 33 loss-making products; operational analysis established a 12.07% return rate driven mainly by size and damage issues and confirmed UPI as the dominant payment method; and review analysis showed 59.7% positive sentiment. Delivering all of these insights through one dashboard demonstrates how unified analytics supports data-driven retail decisions.

B. Future Work

- Supervised churn prediction and Customer Lifetime Value forecasting.
- Time-series sales forecasting and market-basket analysis for cross-selling.
- Transformer-based aspect-level sentiment analysis of review text.
- Streaming ingestion (Apache Kafka or Airflow) and a PostgreSQL or Snowflake data warehouse.
- Containerised cloud deployment with authentication and role-based access.

References

[1] E. W. T. Ngai, L. Xiu, and D. C. K. Chau, "Application of data mining techniques in customer relationship management: A literature review and classification," *Expert Systems with Applications*, vol. 36, no. 2, pp. 2592–2602, 2009.

[2] D. Chen, S. L. Sain, and K. Guo, "Data mining for the online retail industry: A case study of RFM model-based customer segmentation using data mining," *Journal of Database Marketing & Customer Strategy Management*, vol. 19, no. 3, pp. 197–208, 2012.

[3] A. Griva, C. Bardaki, K. Pramatari, and D. Papakiriakopoulos, "Retail business analytics: Customer visit segmentation using market basket data," *Expert Systems with Applications*, vol. 100, pp. 1–16, 2018.

[4] R. Agrawal and R. Srikant, "Fast algorithms for mining association rules," in *Proc. 20th Int. Conf. Very Large Data Bases (VLDB)*, 1994, pp. 487–499.

[5] A. M. Hughes, *Strategic Database Marketing*. Chicago, IL: Probus Publishing, 1994.

[6] C.-H. Cheng and Y.-S. Chen, "Classifying the segmentation of customer value via RFM model and RS theory," *Expert Systems with Applications*, vol. 36, no. 3, pp. 4176–4184, 2009.

[7] D. Birant, "Data mining using RFM analysis," in *Knowledge-Oriented Applications in Data Mining*, K. Funatsu, Ed. Rijeka, Croatia: InTech, 2011, pp. 91–108.

[8] J. B. MacQueen, "Some methods for classification and analysis of multivariate observations," in *Proc. 5th Berkeley Symp. Math. Statist. Prob.*, vol. 1, 1967, pp. 281–297.

[9] A. K. Jain, "Data clustering: 50 years beyond K-means," *Pattern Recognition Letters*, vol. 31, no. 8, pp. 651–666, 2010.

[10] P. J. Rousseeuw, "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis," *Journal of Computational and Applied Mathematics*, vol. 20, pp. 53–65, 1987.

[11] L. Kaufman and P. J. Rousseeuw, *Finding Groups in Data: An Introduction to Cluster Analysis*. New York, NY: Wiley, 1990.

[12] T. Zhang, R. Ramakrishnan, and M. Livny, "BIRCH: An efficient data clustering method for very large databases," *ACM SIGMOD Record*, vol. 25, no. 2, pp. 103–114, 1996.

[13] S. Guha, R. Rastogi, and K. Shim, "CURE: An efficient clustering algorithm for large databases," *Information Systems*, vol. 26, no. 1, pp. 35–58, 2001.

[14] M. Wedel and W. A. Kamakura, *Market Segmentation: Conceptual and Methodological Foundations*, 2nd ed. Boston, MA: Kluwer Academic, 2000.

[15] D. J. Power, *Decision Support Systems: Concepts and Resources for Managers*. Westport, CT: Quorum Books, 2002.

[16] S. Few, *Information Dashboard Design: Displaying Data for At-a-Glance Monitoring*, 2nd ed. Burlingame, CA: Analytics Press, 2013.

[17] W. McKinney, "Data structures for statistical computing in Python," in *Proc. 9th Python in Science Conf. (SciPy)*, 2010, pp. 56–61.

[18] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[19] Plotly Technologies Inc., "Plotly Python graphing library." [Online]. Available: https://plotly.com/python/

[20] Streamlit Inc., "Streamlit documentation." [Online]. Available: https://docs.streamlit.io

[21] G. Shmueli, P. C. Bruce, P. Gedeck, and N. R. Patel, *Data Mining for Business Analytics: Concepts, Techniques, and Applications in Python*. Hoboken, NJ: Wiley, 2019.

[22] Kaggle, "E-Commerce Sales and Customer Analytics (simulated Indian e-commerce dataset)." [Online]. Available: https://www.kaggle.com

[23] J. Han, J. Pei, and H. Tong, *Data Mining: Concepts and Techniques*, 4th ed. Cambridge, MA: Morgan Kaufmann, 2022.

[24] P.-N. Tan, M. Steinbach, A. Karpatne, and V. Kumar, *Introduction to Data Mining*, 2nd ed. New York, NY: Pearson, 2019.

[25] B. Liu, *Sentiment Analysis and Opinion Mining*. San Rafael, CA: Morgan & Claypool, 2012.

[26] F. Provost and T. Fawcett, *Data Science for Business*. Sebastopol, CA: O'Reilly Media, 2013.